TASK SPECIFICATION: P8-04 PHASE P8 — SECURITY HARDENING, COMPLIANCE & QUALITY ASSURANCE

High-Concurrency Load & Stress Testing (JMeter / k6)

Git Branch: `test/P8-04-performance` · Target Pull Request: test/P8-04-performance

Core Objective & Technical Scope:
Concurrency stress testing of financial ledger transfers and parking spot reservations under simulated peak campus traffic.

Priority Level	Medium	Complexity Rating	`M`
Execution Status	Backlog 📋	Target PR / Commit	`test/P8-04-performance`
Upstream Dependencies	P7-06		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Author load testing scenarios simulating 500 concurrent booking attempts.
2. Execute load tests against containerized PostgreSQL and backend environment.
3. Analyze database lock contention and optimize connection pool parameters.
4. Document latency percentiles (p95, p99) under peak load.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `backend/src/test/**`
- `load scripts`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Zero balance drift or negative balances under maximum concurrency.
- Zero double-bookings permitted by database exclusion constraints.

VERIFICATION & TEST COVERAGE STRATEGY

- JMeter/k6 load test execution
- Database integrity post-load audit